Statistics method to predict Churn

1. Our data

Our dataset contains **7,043 customers** and **21 variables**. The variable we want to predict is **Churn**, which indicates whether a customer left the company (Yes) or stayed (No).

We have different types of information about the customers:

- **Personal information**: gender, SeniorCitizen, Partner, Dependents
- **Customer tenure**: tenure
- **Services used**: PhoneService, InternetService, OnlineSecurity, OnlineBackup, TechSupport, StreamingTV, etc.
- **Contract and payment information**: Contract, PaperlessBilling, PaymentMethod
- **Prices**: MonthlyCharges and TotalCharges

In the dataset, around **73.5% of customers stayed** and **26.5% left the company**.

Our first step would therefore be to identify which variables seem to be related to Churn.

2. Using linear regression

Once we have identified the variables that seem to be important, we could try to use **linear regression** to obtain a prediction of Churn.

We could transform the variable as follows:

Churn = 0 if the customer stays

Churn = 1 if the customer leaves

The regression could then have a form such as:

Estimated Churn = $a + b_1X_1 + b_2X_2 + ... + b_kX_k + e$

For example, the X variables could be tenure, MonthlyCharges, etc.

The idea would be to use the characteristics of a customer to obtain a **predicted value**. A value close to 0 would indicate a customer who is more likely to stay, while a value close to 1 would indicate a customer who is more likely to leave.

We could then compare the predicted values with the actual Churn values to see whether our method works correctly.

Linear regression would therefore be a **first approach**, based on the methods we have already studied. It could help us determine whether the variables in the dataset can actually be used to predict customer behaviour.